

Some review questions - 3rd week

Q1 What condition must be satisfied in order to multiply $A \in \mathbb{R}^{m \times n}$ with $B \in \mathbb{R}^{k \times \ell}$? When this condition holds, give the definition of the entries of $C = AB$ in terms of the entries of A and B .

Q2 Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & -1 \end{pmatrix}.$$

Which of the two products AB and BA are defined? Compute the product(s) whenever possible.

Q3 Given $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times \ell}$, let us denote by $a_{(1)}^\top, \dots, a_{(m)}^\top$, the rows of A and by $b_1, \dots, b_\ell$, the columns of B . For $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, \ell\}$:

- (a) How many additions and multiplications are required to compute $a_{(i)}^\top b_j$?
- (b) How many operations are required to compute Ab_j ?
- (c) How many operations are required to compute AB ?

Your answer may be an approximation of the exact result when m, n, ℓ are large (for instance, when n is large, you may approximate $n - 1$ by n).

Q4 Write explicitly the identity matrix and the zero matrix in $\mathbb{R}^{3 \times 3}$. What properties do they satisfy with respect to matrix multiplication?

Q5 Using appropriate dimensions, state precisely the following properties of matrix multiplication:

- (a) left and right distributive rules;
- (b) associativity with respect to scalar multiplication;
- (c) associativity of matrix multiplication.

Q6 Prove that the matrix–vector product and the matrix–matrix product are left distributive, i.e.

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \quad A(B + C) = AB + AC,$$

for matrices A, B, C and vectors $\mathbf{u}, \mathbf{v}$ of compatible sizes.

Q7 Give an example of two square matrices $A, B \in \mathbb{R}^{n \times n}$ such that $AB = BA$, and an example such that $AB \neq BA$. Which fundamental algebraic property does matrix multiplication fail to satisfy?

Q8 Give the general definition of the transpose of a matrix $A \in \mathbb{R}^{n \times m}$. Compute the transpose of

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \end{pmatrix}.$$

Q9 Define symmetric matrices. Give one example of a symmetric matrix and one example of a matrix that is not symmetric.

Q10 Given $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$, express $(AB)^T$ as the product of two matrices and prove the corresponding identity.

Q11 Let $\mathbf{u}, \mathbf{v}$ be column vectors of the same size; more precisely, assume $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{n \times 1}$.

- (a) Express the inner product and the outer product of $\mathbf{u}$ and $\mathbf{v}$ using matrix multiplication and transpose.
- (b) Are the inner products $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$ the same? Are the outer products $\mathbf{u}\mathbf{v}^T$ and $\mathbf{v}\mathbf{u}^T$ the same? Briefly justify.
- (c) If $\mathbf{w} \in \mathbb{R}^p$ with $p \neq n$, can the inner product of $\mathbf{u}$ and $\mathbf{w}$ be defined? What about the outer product of $\mathbf{u}$ and $\mathbf{w}$?

Q12 Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C = (3), \quad D = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}.$$

Construct a block matrix M that contains exactly these four matrices as blocks. Give one expression using block notation with A, B, C, D , and one expression written explicitly in scalar entries.

Q13 Give two conditions for a block partition

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{pmatrix}$$

to be valid.

Q14 Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

be two matrices with valid block partitions. Give two sufficient conditions ensuring that the product AB is well-defined. Assuming these conditions hold, express AB in block form.

Q15 Fill in the following table whenever possible. Identify which entries correspond to inner products and which correspond to outer products.

	B	$\begin{pmatrix} b^{(1)} \\ b^{(2)} \\ b^{(3)} \end{pmatrix}$	$\begin{pmatrix} b_1 & b_2 \end{pmatrix}$
A	AB		
$\begin{pmatrix} a^{(1)} \\ a^{(2)} \end{pmatrix}$			
$\begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix}$			